

Tutorial No : 3

Tutorial Title: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

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Batch: C

Program:

```
#include <iostream>
using namespace std;
class arithmetic
{
    int r1, i1, r2, i2;
    float rr, ii;
public:
    void getdata(int a, int b, int c, int d);
    void putdata();
};
void arithmetic::getdata(int a, int b, int c, int d){
    r1 = a;
    i1 = b;
    r2 = c;
    i2 = d;
}
void arithmetic::putdata(){
    cout << "First Complex Number = " << r1 << " + " << i1 << "i" << endl;
    cout << "Second Complex Number = " << r2 << " + " << i2 << "i" << endl;
    cout << "\nAddition = "
        << (r1 + r2) << " + " << (i1 + i2) << "i";
    cout << "\nSubtraction = "
        << (r1 - r2) << " + " << (i1 - i2) << "i";
    cout << "\nMultiplication = "
```

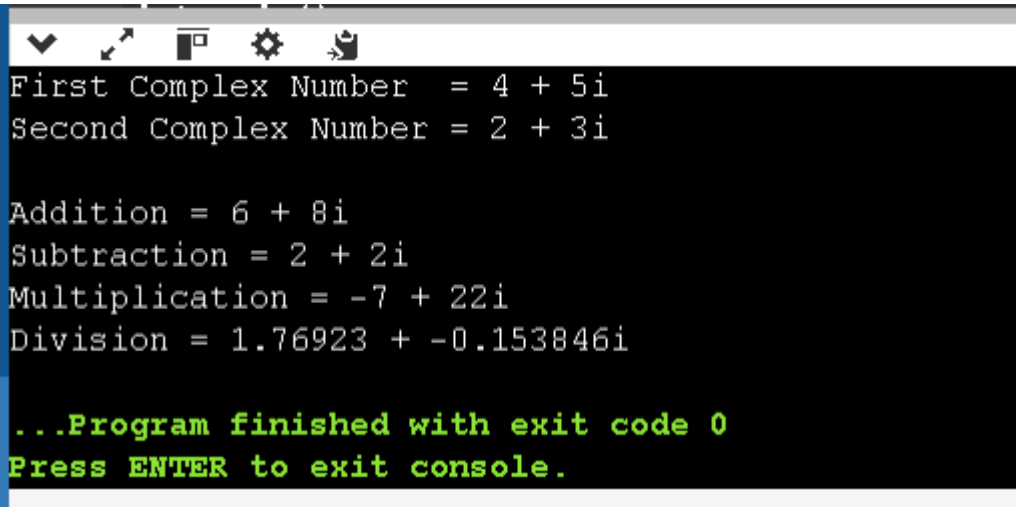
```

        << (r1 * r2 - i1 * i2) << " + "
        << (r1 * i2 + i1 * r2) << "i";
float den = r2 * r2 + i2 * i2;
if (den != 0) {
    rr = (float)(r1 * r2 + i1 * i2) / den;
    ii = (float)(i1 * r2 - r1 * i2) / den;

    cout << "\nDivision = "
        << rr << " + " << ii << "i";
}
else {
    cout << "\nDivision not possible.";
}
}
int main()
{
    arithmetic a;
    a.getdata(4, 5, 2, 3);
    a.putdata();
    return 0;
}

```

Output:



```

First Complex Number = 4 + 5i
Second Complex Number = 2 + 3i

Addition = 6 + 8i
Subtraction = 2 + 2i
Multiplication = -7 + 22i
Division = 1.76923 + -0.153846i

...Program finished with exit code 0
Press ENTER to exit console.

```